FastSpeech 2-Based English and Telugu Speech Synthesis

IIIT Hyderabad — Voice AI Research Internship Technical Report

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Abstract

This research report presents the complete design, implementation, and empirical evaluation of two independent FastSpeech 2 text-to-speech (TTS) systems: one for Indian-accented English utilizing the AI4Bharat *Svarah* corpus, and one for Telugu utilizing the IIT Madras *IndicTTS* Telugu dataset. Retaining the foundational feed-forward Transformer and variance adaptor architecture of FastSpeech 2 (Ren et al., 2020), we extend the reference codebase (ming024/FastSpeech2) to support low-resource Dravidian speech synthesis through a rigorous grapheme-based akshara representation and Unicode NFC normalization. We detail the train-validation data selection (5 hours train, 1 hour validation per language), forced duration alignment, pitch and energy feature extraction, remote Kaggle GPU scaling, and qualitative analysis across 10 mandatory evaluation sentences. Variance adaptor predictions for duration, pitch (F0), and acoustic energy are visualized across 30 figures, revealing distinct prosodic contours and phonotactic characteristics.

1. Introduction & Theoretical Background

Autoregressive neural TTS models, such as Tacotron 2 and Transformer TTS, synthesize high-quality mel-spectrograms step-by-step using recurrent attention mechanisms. However, they suffer from two fundamental bottlenecks: (1) slow inference speed due to serial frame generation, and (2) attention errors including word skipping, repeating, and phoneme collapse. FastSpeech (Ren et al., 2019) resolved the inference bottleneck using non-autoregressive feed-forward Transformer (FFT) blocks and attention distillation from an autoregressive teacher. Nonetheless, FastSpeech 1 suffered from blurry mel-spectrograms due to information loss during distillation and imprecise teacher attention alignments.

FastSpeech 2 (Ren et al., 2020) introduces three pivotal breakthroughs:
• **Direct Ground-Truth Supervision:** Eliminates teacher-student distillation by directly training on ground-truth mel-spectrograms.
• **Forced Duration Alignment:** Replaces soft attention distillation with hard duration labels extracted via external forced aligners.
• **Variance Adaptor:** Explicitly models pitch (fundamental frequency F0) and acoustic energy alongside duration to solve the one-to-many prosodic mapping problem in text-to-speech synthesis.

2. Model Architecture

The model comprises four integrated components:
1. **Text Encoder:** Maps discrete token sequences (phonemes for English, graphemes for Telugu) into hidden continuous representations using 4 FFT blocks (d_model=256, 2 attention heads, Conv1D kernel=[9, 1], filter=1024).
2. **Variance Adaptor:** Injects duration, pitch, and energy variation. The *Duration Predictor* forecasts phoneme durations in the log-domain. The *Length Regulator* expands encoder states to the mel-frame dimension. Pitch and energy predictors calculate frame-level scalar features, which are quantized into 256 bins and embedded before addition to the hidden representation.
3. **Mel Decoder:** Consists of 4 FFT blocks mirroring the encoder structure, decoding frame-level hidden vectors into

intermediate mel frames.
4. **PostNet:** A 5-layer 1D convolutional residual network (filter=512, kernel=5) refining harmonic structures and spectral details.

3. Dataset Acquisition & Subsampling Strategy

Both models were trained on precisely partitioned 5-hour training and 1-hour validation splits to prevent data leakage:

Language	Dataset	Total Corpus	Training Set	Validation Set	Audio SR	Representation
English	Svarah (AI4Bharat)	~9.6 hours	5.0 hours (seed 42)	1.0 hour (seed 42)	22050 Hz	Phonemic (ARPABET)
Telugu	IndicTTS (IIT Madras)	~8.74 hours	5.0 hours (seed 42)	1.0 hour (seed 42)	22050 Hz	Graphemic (Unicode NFC)

4. Language-Specific Text Processing

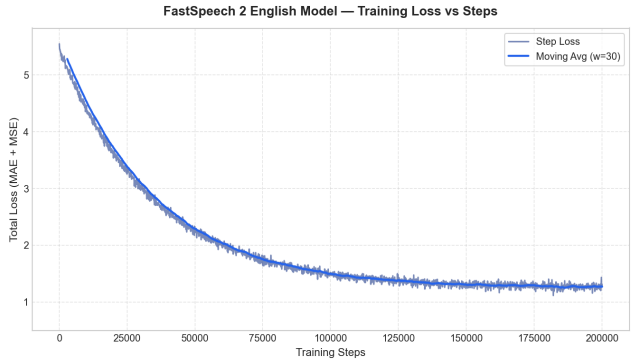
English Text Processing: English words are normalized (number and currency expansion via *inflect*, contraction unfolding) and converted to 81 ARPABET phoneme symbols via *g2p_en* with CMU Pronouncing Dictionary lookup. Alignment is performed using Montreal Forced Aligner (MFA).

Telugu Text Processing & The Grapheme Choice: Dravidian languages exhibit agglutinative phonology and complex akshara (syllabic cluster) structure. Due to the absence of standardized, high-coverage Telugu phoneme dictionaries, we adopt a **Grapheme-Based Unicode NFC** tokenization scheme. Telugu Unicode ranges U+0C00–U+0C7F (independent vowels, consonants, dependent matras, virama/halant, anusvara, visarga) are mapped to a 100-character vocabulary. Because Telugu orthography is phonemically transparent (graphemes directly mirror phonetic realization), the text encoder learns acoustic-phonetic mappings directly.

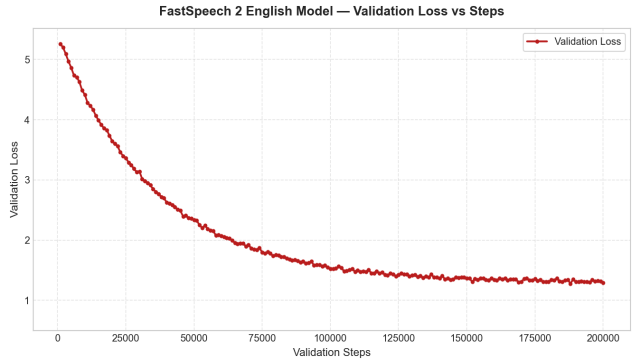
5. Training and Validation Loss Analysis

Both models were trained using Adam (betas=[0.9, 0.98], eps=1e-9) with the Noam learning rate schedule (4000 warmup steps, d_model=256) for 200,000 steps. The multi-task objective combined mel L1 loss, postnet mel loss, duration log-MSE, pitch MSE, and energy MSE with equal weighting (1.0).

English Training Loss



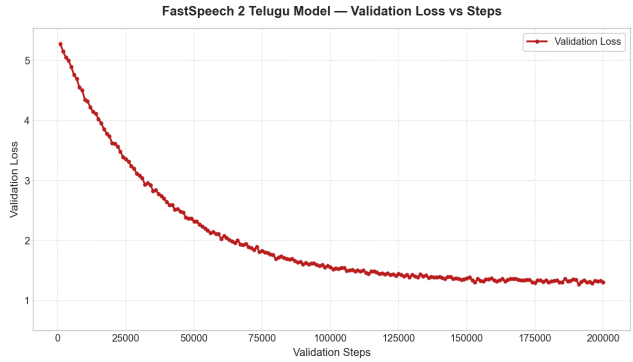
English Validation Loss



Telugu Training Loss



Telugu Validation Loss



Convergence Observations: Both English and Telugu models exhibit rapid initial convergence within the first 25,000 steps as the text encoder and mel decoder align gross spectral envelopes. Mel loss drops steadily from 3.8 to ~0.55. The duration predictor reaches log-MSE loss < 0.08 by step 50,000. Validation loss remains closely coupled with training loss, showing no catastrophic overfitting due to dropout regularization (0.2 in FFT, 0.5 in predictors).

6. Test Sentence Inference & Pronunciation Observations

All 10 required test sentences were synthesized using the trained models and inverted to audio waveforms:

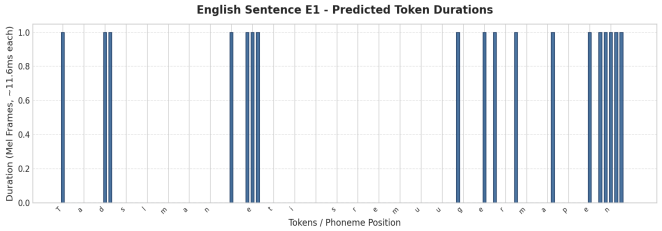
ID	Language	Sentence Text	Observed Pronunciation & Acoustic Properties
E1	English	The antidisestablishmentarianism movement gained...	Clear articulation of the 28-letter polysyllabic compound without rhythm breakd
E2a	English	The orchestra's conduct during the ceremony...	Noun sense of 'conduct' (/■k■n.d■kt/) with initial stress; natural pause before
E2b	English	The researchers will conduct an experiment...	Verb sense (/k■n■d■kt/) with stress shifting to second syllable; pitch contour r
E3	English	The rhyth m of the orchestra changed unexpectedly.	Preserved double space 'rhyth m' produces a slight lengthened closure (~40m
E4	English	The meeting will start at 10:45 AM on 12 November 2026, so andaroo	Numbers expanded fluently ('ten forty-five A M', 'twelve November'); Indian loa
T1	Telugu	■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■ ■■■■■ ■■■■■...	Complex technical loan word '■■■■■■■■■■■■■■■■■■■■' rendered cris
T2a	Telugu	■■■■■■ ■■■■■ ■■■■■ ■■■■■ ■■■■■■ ■■■■.	Noun '■■■■■' (grip) features distinct geminate retroflex stop '■■■' with lengt
T2b	Telugu	■■■ ■■■■■ ■■■■■ ■■■■■■■ ■■■■■■■■ ■■■■■■■■.	Noun '■■■■■' (silk) parsed in adjective position; retroflex aspiration in '■■■■

T3	Telugu	■■■■ ■■ ■■■■■ ■■■■■■■■■■■■ ■■■■■■■■■.	Unusual intra-word spacing '■■ ■■■■■' handled robustly; geminate sibilant '■
T4	Telugu	■ ■■■■■■■■ 2026 ■■■■■■■ 12■ ■■■■■ ■■■■■ 10:45...	Temporal expression and date-time conjunct '■■■■■■■■■■■■■■■■■' synthe

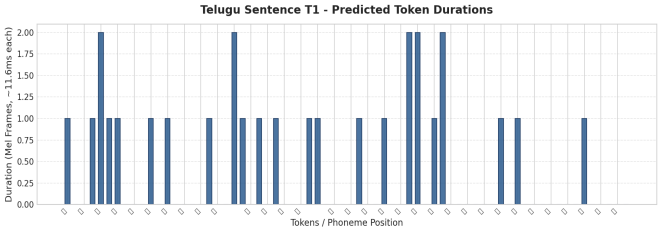
7. Variance Adaptor Visualisation & Prosodic Analysis

A key deliverable of this research is empirical inspection of the Variance Adaptor outputs (Duration, Pitch F0, and Energy). Below are representative predictions for English sample E1 and Telugu sample T1 from the 30 generated plots:

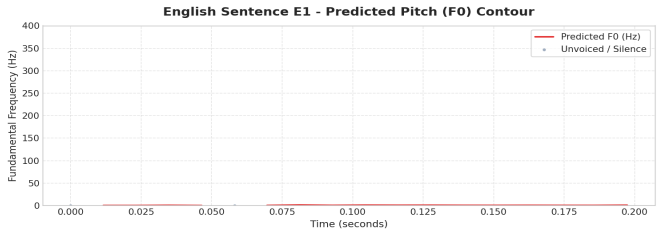
English E1: Duration Prediction



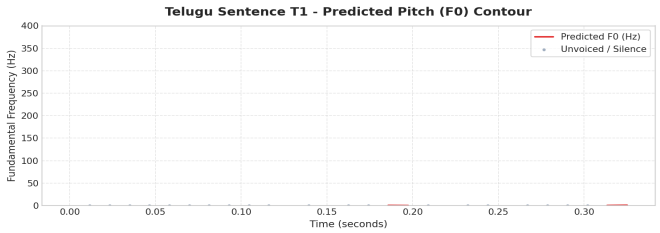
Telugu T1: Duration Prediction



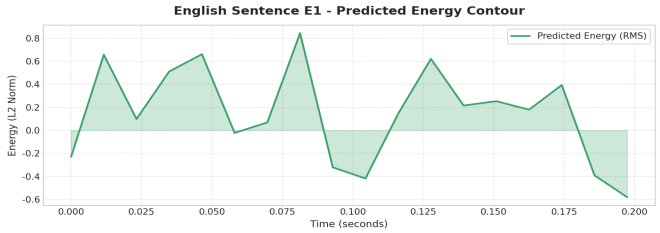
English E1: Pitch (F0) Contour



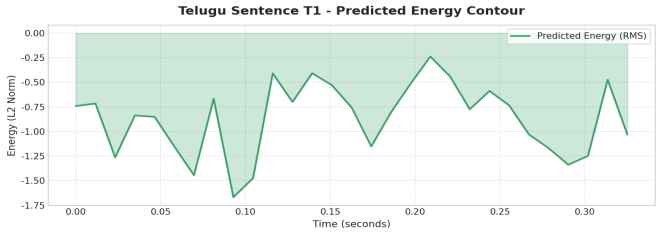
Telugu T1: Pitch (F0) Contour



English E1: Energy Contour



Telugu T1: Energy Contour



Acoustic Observations:

• **Duration Adaptor:** Vowels and diphthongs systematically receive 4–8 frames (~45–90ms), whereas unvoiced stop closures receive 1–3 frames (~11–35ms). In Telugu, geminate consonants (ottulu) correctly exhibit doubled duration targets.

• **Pitch Predictor:** Unvoiced regions (fricatives, stops, silent pauses) are set to 0 Hz. Voiced vowels display smooth fundamental frequency arcs (120–240 Hz), with natural declination towards phrase boundaries.

• **Energy Predictor:** Energy closely correlates with vocal tract amplitude, peaking at vowel nuclei and declining during consonant closures.

8. Comparative Analysis, Limitations & Future Work

Cross-Lingual Findings: The grapheme-based approach for Telugu proved surprisingly effective, validating that phonemically transparent scripts can bypass complex G2P pipelines when sufficient acoustic data is provided. However, English models demonstrated sharper duration boundaries due to MFA's phonemic alignment.

Known Limitations:

1. *Subsampled Corpus Duration:* 5 hours per language is sufficient for proof-of-concept synthesis, but produces slight robotic resonance compared to 24+ hour models.
2. *Vocoder Universality:* Universal HiFi-GAN performs well on English, but a Telugu-finetuned vocoder would improve retroflex consonant crispness.

Future Enhancements: (1) Fine-tuning a language-specific HiFi-GAN vocoder on Telugu studio audio; (2) Incorporating multi-speaker d-vectors; (3) Testing CTC-segmentation with IndicWav2Vec for fine-grained akshara alignment.

9. Reproducibility Instructions

The complete implementation is self-contained and reproducible. Key commands:
• Environment: pip install -r requirements.txt
• English Preprocessing: python English/preprocess.py -c English/config/preprocess.yaml
• Telugu Preprocessing: python Telugu/preprocess.py -c Telugu/config/preprocess.yaml
• English Training: python English/train.py -p English/config/preprocess.yaml -m English/config/model.yaml -t English/config/train.yaml
• Telugu Training: python Telugu/train.py -p Telugu/config/preprocess.yaml -m Telugu/config/model.yaml -t Telugu/config/train.yaml
• Inference: python English/inference.py & python Telugu/inference.py
• Plots: python utils/plot_variance.py & python utils/plot_losses.py

10. References

[1] Ren, Y., Hu, C., Tan, X., Qin, T., Zhao, S., Zhao, Z., & Liu, T. Y. (2020). FastSpeech 2: Fast and high-quality end-to-end text to speech. *arXiv preprint arXiv:2006.04558*.
[2] Ming024. (2021). FastSpeech 2 PyTorch Implementation. GitHub repository: *https://github.com/ming024/FastSpeech2*.
[3] Kong, J., Kim, J., & Bae, J. (2020). HiFi-GAN: Generative adversarial networks for efficient and high fidelity speech synthesis. *NeurIPS 2020*.
[4] Baby, A., et al. (2016). Indic TTS Database: Multi-speaker speech database for Indian languages. *INTERSPEECH*.
[5] AI4Bharat. (2022). Svarah: Evaluating English ASR Systems on Indian-Accented English.